Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 1 of 10 · Question 5 part c · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

5 (a)	(i)	• flows from artery; • where blood is at high pressure; • into vein; • because blood is at a low pressure;	mp1- accept in context of artery leaving heart mp3- accept in context vein of returning to the heart If comparative answer given for pressure score marking point 2 and 4	4
	(ii)	glomerulus;		1
	(iii)	• large (surface) area; • for diffusion / molecules/ named molecules to pass out(of blood);	More area for diffusion = 2 marks	2
	(iv)	more urea;		1
(b)		• no/less gradient between liquid X and blood (glucose)/blood and X have same concentration/eq; • glucose does not pass out/ glucose is not lost: • by diffusion • needed for respiration; • minimises osmotic effect;		3
(c)		• protein digested/broken down into amino acids; • (excess) **amino acids** broken down/deaminated; • urea; • less protein means less urea produced; • less urea means less to be filtered/removed ; • high concentration of urea/nitrogenous waste can be toxic;	MP3– accept urea from protein breakdown	4
				Total =15

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 1 of 10 · Question 5 part c · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 2 of 10 · Question 8 part a.i · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

8(a)(i)	broken down in liver; by deamination; to form urea; used to make new protein/amino acids;		**(3)**

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	poor/restricted movement of synovial joints pain in joint; backflow of blood in heart; poor/inefficient blood circulation/oxygenation; less energy/tired;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
8(b)(i)	• not sex-linked; • more females affected than males/more males would be affected; • only females would be carriers (if sex-linked); • male 1/3/11 would show condition if sex-linked; • but these males are carriers;		**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	• probability that child has condition is ¼ /1:3 / 1 in 4/25%/0.25; • probability that child is male is ½/1 in 2/1:1 / 50%/0.5; • so probability that it is affected male is 12.5%/0.125/$\frac{1}{8}$;		**(3)**

Total 13 Marks

Paper Total 90 Marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 2 of 10 · Question 8 part a.i · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 3 of 10 · Question 9 part a.i · 3 marks

2020 JAN · Paper 02 · find part a.i in this mark scheme extract

Number					
9	(a)	(i)	Any three from: • amino group removed/broken down(1) • in liver(1) • by deamination(1) • to produce urea (1)	Allow description of amino group e.g. nitrogenous group	3
		(ii)	Any three from: • less protein broken down by protease(1) • to amino acids 1) • in stomach/small intestine(1) • reference to reduced amount of phenylalanine/amino acids (in low protein diet)(1)		3
		(iii)	• reduced number of amino acids available(1) • during translation(1) • on ribosomes(1) • incorrect/less protein formed(1)	Allow less translation Allow less protein synthesis	3
	(b)	(i)	heterozygous/Nn(1)		1
		(ii)	100% (1)		1
		(iii)	• both parents carry recessive allele/allele for PKU/heterozygous(1) • person 3 has inherited one recessive allele from each parent(1) • person 3 has PKU/homozygous recessive(1)	Reject any reference to sex-linked	3

Total for Question 9 = 14 marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 4 of 10 · Question 7 part b.i · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
7	(a)	(i)	Hypothalamus/osmoregulatory centre;		1
		(ii)	<table><tr><th></th><th colspan="3">Structure</th></tr><tr><th>Function</th><th>glomerulus</th><th>proximal tubule</th><th>renal vein</th></tr><tr><td>regulation of water level</td><td></td><td></td><td></td></tr><tr><td>ultrafiltration</td><td>✓</td><td></td><td></td></tr><tr><td>reabsorption of glucose</td><td></td><td>✓</td><td></td></tr><tr><td>removal of blood from kidney</td><td></td><td></td><td>✓</td></tr><tr><td>supply of blood to kidney</td><td></td><td></td><td></td></tr><tr><td>removal of urine from collecting duct</td><td></td><td></td><td></td></tr></table>	Reject more than one tick in any column	3
	(b)	(i)	• breakdown of proteins/amino acids/deamination/removal of NH_2/amine group; • in the liver;		2
		(ii)	Five from: • second sample has less water/more urea/other components; • urine is more concentrated/less urine produced; • person is less hydrated/less water/lower water potential in the blood; • ADH has been produced; • increased the permeability of the collecting duct; • more water has been reabsorbed (by the blood);		5

Total 11 marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 4 of 10 · Question 7 part b.i · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 5 of 10 · Question 7 part b · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number					
7	(a)	(i)	• axes labelled with units; • correct way round and suitable scale; • correct plots; • suitable line drawn;		4
		(ii)	• only not working correctly when rate falls below 100; • data/year not shown when that occurs/already below 100;		2
		(iii)	• 2021/2022; • rate falling at 5-10 units per annum/similar values each year;		2
		(iv)	• not very accurate/results vary/fluctuate; • earlier data showed improvement in function; • could happen again/results may change;		3
	(b)		any three from • protein broken down into urea/deamination; • toxic; • needs to be excreted; • incorrectly functioning kidney can't remove it all if there is a lot; • less protein less urea;		3

Total 14

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 5 of 10 · Question 7 part b · 3 marks

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Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 6 of 10 · Question 7 part c.ii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.ii in this mark scheme extract

7 (a) (i)	C; (ureter) *A is a blood vessel* *B is a blood vessel* *D is not attached to kidney*		1
(ii)	A; (bladder) *B is part of the reproductive system* *C is part of the reproductive system* *D leads from the bladder*		1
(b)	• more oxygen; • less carbon dioxide; • more urea;	accept reverse for blood leaving kidney	3
(c) (i)	16.8 - 3.6 = 13.2; 13.2 ÷ 3.6 =3.66; 3.66 × 100 = 366.7%;		3
(ii)	• urea; • large/significant increase in protein-rich diet;		2
(iii)	Any four from • urea, uric acid and ammonia masses are different between the two diets; • ref. creatinine; • amount same/nearly same; • in both diets; • possible result of muscle breakdown;		4

Total 14 marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 6 of 10 · Question 7 part c.ii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.ii in this mark scheme extract

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Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 7 of 10 · Question 8 part b · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

8(a)	<table><tr><th rowspan="2">Organ</th><th rowspan="2">Mass in kg</th><th colspan="2">Rate of blood flow in cm^3 per minute</th></tr><tr><th>through the whole organ</th><th>per 100g of the organ</th></tr><tr><td>kidneys</td><td>0.3</td><td>1200</td><td>400</td></tr><tr><td>skeletal muscles</td><td>25.0</td><td>750</td><td>3</td></tr><tr><td>liver</td><td>1.5</td><td>1500</td><td>100</td></tr></table> kidney 1200÷3; 400; skeletal muscle 250 × 3; 750;	correct answer award full marks Ecf for each calculation allow correct answer to wrong figures	**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)	Any four from: • (receives blood from) hepatic artery; • (receives blood from) hepatic portal vein; • homeostasis; • reference to blood sugar regulation; • detoxification (of alcohol)/urea formation/deamination;	For mp 1 and 2 allow receives blood from two vessels for one mark	**(4)**

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 7 of 10 · Question 8 part b · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
8(c)	Any four from: • less blood flow to kidney/liver; • increased blood flow to skeletal muscles; • (muscles have) greater energy demand/more energy needed/to reduce oxygen debt/lactic acid; • (so) more oxygen/glucose needed; • for (aerobic) respiration;		**(4)**

Total 12 Marks

Paper Total 90 Marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 8 of 10 · Question 4 part b.i · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

4 (a)	• reduce fat/cholesterol intake; • increase exercise;		2
(b) (i)	Any four from • liver breaks down toxins/drugs; • (if not working properly) statins won't be broken down/will accumulate/reduced detoxification; • kidneys excrete waste (products)/named waste products; • (if not working) can't be removed/less waste/named waste filtered/removed; • increased risk of (further) liver/kidney damage;	Ignore references to deamination	4
(ii)	• statins could cross from mother to fetus; • through the placenta; • damage/affect growth/development of fetus/lead to low birth weight of fetus; • could be in mother's milk if breastfeeding;	Allow baby for mp's 1 and 3 Ignore affects fetus	3
(c) (i)	• (16.3 : 8) = 2.0;	Accept 2.0375 Accept 2/2.04	1
(ii)	Any four from • Double the number/more symptoms shown with patients taking statins than those taking no tablets • even on statins symptoms not high; • almost same with placebo/symptoms with placebo similar/just below those on statins; • symptoms still present when taking no statins; • statins not the cause/other causes of symptoms/named cause e.g. age/gender/fitness/lifestyle;		4 exp

Total 14 marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 8 of 10 · Question 4 part b.i · 4 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks
			2

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 9 of 10 · Question 6 part a.ii · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

6	(a)	(i)	any two from: • alcohol absorbed/passes into blood; • rapidly/immediately/straight away/steep increase; • by stomach;	ignore intestines	2
		(ii)	• (alcohol carried to) liver; • broken down/detoxified; • level (in blood) falls;		3
	(b)	(i)	• any time after (including) 2.4 hours/2 hours 24 mins;	allow a time of between 1-20 minutes	1
		(ii)	• alcohol is a depressant; • slows reaction/increases reaction time; • drivers can't respond as quickly/less alert; • leads to accidents;		4
	(c)		• collecting duct less permeable (in absence of ADH); • less water reabsorbed; • into blood; • volume of urine (therefore) increased/more urine produced; • and is more dilute/less concentrated;	accept ref fewer aquaporins/water channels	5
					15 marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 9 of 10 · Question 6 part a.ii · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 10 of 10 · Question 5 part e · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

5 (a)	• ACE inhibitor prevents/inhibits action of enzyme; • prevents formation of angiotensin II; • less(angiotensin ii) bind with receptors; • Losartan blocks/binds to receptors; • angiotensin ii cannot bind to receptor;		4
(b)	• arteries/arterioles; • muscles in walls; • can contract/constrict; • narrow lumen; • smaller volume/space(for blood flow);		4
(c)	• in normal/healthy kidney protein not found in urine; • protein is a big molecule; • should not pass through the glomerulus/basement membrane;		2
(d)	• damage to glomerulus/basement membrane; • stronger/intense/eq; <u>contractions</u>; • (which causes) <u>muscle</u> to get thicker;	mp2 must be due to mp1	2
(e)	• detoxification occurs in the liver; • can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver; • results in (further) liver damage/liver failure/liver diseae;		2 **Total =14**

Type MS · Explain the liver's roles in deamination and breaking down drugs

Answer 10 of 10 · Question 5 part e · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks
	• A=rods/cones/retina;		